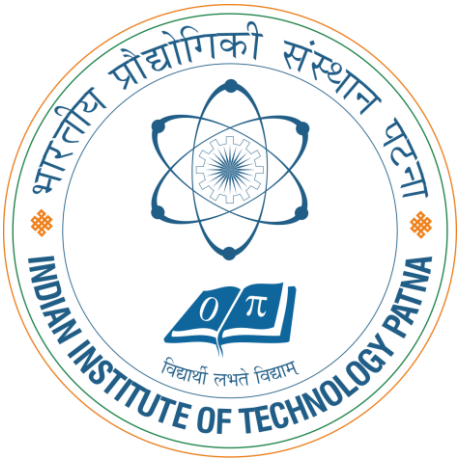
Report of Assignment – 4

Course: MA6106 (LLM)



Indian Institute of Technology Patna

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Overview

This report presents a combined analysis of two sequence-modelling experiments using recurrent neural networks (RNNs). Task 1 treats each 28 × 28 image of the Kannada-MNIST Dig-MNIST set as a sequence of 28 row vectors and compares a vanilla RNN, an LSTM and a GRU on ten-class handwritten digit recognition, using Keras with the Adam optimizer and early stopping. Task 2 classifies spoken Arabic digits from variable-length MFCC sequences with a PyTorch implementation, compares RNN, LSTM and GRU cells with one or two layers and hidden sizes of 32, 64 and 128 (18 experiments), and then studies bidirectional and attention-based extensions together with gradient-flow diagnostics.

In Task 1 the selected model, a 64-unit GRU, obtains 91.15% validation accuracy and 89.58% test accuracy. In Task 2 the selected baseline, a two-layer 32-unit GRU, obtains 99.55% validation accuracy and 96.27% test accuracy; among all models including the extensions, a bidirectional two-layer GRU has the highest validation accuracy (99.70%) and reaches 97.55% on the test set. In both tasks the gated cells (LSTM and GRU) reached higher validation accuracy than the vanilla RNN; in Task 2 they also trained more reliably and generalized clearly better.

TASK 1: Kannada Digit Classification with RNN, LSTM and GRU

1. Objectives

- Represent each 28 × 28 digit image as a sequence of 28 time steps with 28 features per step.
- Apply a stratified 70/15/15 train-validation-test split.
- Implement vanilla RNN, LSTM and GRU classifiers for ten digit classes.
- Compare six hyperparameter configurations (hidden units, dropout and learning rate).
- Plot training and validation error against epochs for every configuration.
- Select the best configuration using validation accuracy.
- Evaluate the selected model on the held-out test set using accuracy, a classification report and a confusion matrix.
- Compare the three recurrent architectures and visualize the row-wise sequential input.

2. Dataset and Preprocessing

2.1 Kannada-MNIST (Dig-MNIST) Dataset

The data were read from the Dig-MNIST file of the Kannada-MNIST collection. The file contains 10,240 labelled samples, each with a digit label and 784 pixel columns (a 28 × 28 grayscale image), covering the ten Kannada digit classes 0–9. Pixel intensities were scaled to the interval [0, 1] by dividing by 255. Figure 1 shows five sample images of every class.

2.2 Sequential Input Representation

Each image is reshaped into a sequence of 28 time steps, where time step t is the t-th row of the image and each step carries 28 pixel features. The network input therefore has shape (samples, 28, 28). The recurrent layer reads the image from the top row to the bottom row, and its final hidden state summarizes the whole digit.

2.3 Data Split

A stratified split with random seed 42 was used: 70% of the samples for training, then the remaining 30% divided equally into validation and test sets. Stratification keeps the class proportions the same in every subset; the test set holds 153 or 154 samples per class.

Subset	Samples	Percentage	Input shape
Training	7,168	70%	(7168, 28, 28)
Validation	1,536	15%	(1536, 28, 28)
Test	1,536	15%	(1536, 28, 28)
Total	10,240	100%	—



Figure 1. Five sample images for each Kannada digit class.

3. Methodology

3.1 Recurrent Network Design

All models share the same structure: an input of shape 28 × 28, one recurrent layer (SimpleRNN, LSTM or GRU) that returns only its final hidden state, a fully connected layer

with 64 ReLU units, and a 10-unit softmax output layer. Only the type and size of the recurrent layer, its dropout rate and the learning rate differ between configurations. The dropout argument of the recurrent layer is applied to its input connections.

3.2 Loss Function and Optimizer

The models are trained with sparse categorical cross-entropy, which compares the integer digit label with the softmax output. Optimization uses the Adam optimizer with the learning rate specified for each configuration, a mini-batch size of 128 and a maximum of 20 epochs. Classification error (1 − accuracy) is recorded after every epoch for both the training and the validation sets.

3.3 Hyperparameter Configurations

Six configurations were tested, two for each recurrent cell type. The larger model of each pair uses 128 units with a dropout rate of 0.2, and the larger LSTM and GRU also use a smaller learning rate of 0.0005.

Configuration	Cell	Units	Dropout	Learning rate	Parameters
RNN_64_lr001	RNN	64	0.0	0.001	10,762
RNN_128_lr001	RNN	128	0.2	0.001	29,002
LSTM_64_lr001	LSTM	64	0.0	0.001	28,618
LSTM_128_lr0005	LSTM	128	0.2	0.0005	89,290
GRU_64_lr001	GRU	64	0.0	0.001	22,858
GRU_128_lr0005	GRU	128	0.2	0.0005	69,578

3.4 Training Procedure and Early Stopping

A fixed seed of 42 was used for Python, NumPy and TensorFlow, and the model was rebuilt from scratch for every configuration. Training used early stopping on the validation loss with a patience of four epochs and restored the weights from the epoch with the lowest validation loss. The reported training, validation and test accuracies are computed with these restored weights.

4. Experimental Results and Analysis

4.1 Overall Training, Validation and Test Performance

The six configurations reach validation accuracies between 84.90% and 91.15%. The GRU with 64 units ranks first on validation accuracy, followed by the 128-unit LSTM and the 64-unit LSTM. The vanilla RNN with 64 units is clearly the weakest model.

Configuration	Epochs	Train Acc.	Val. Acc.	Test Acc.
GRU_64_lr001	**20**	**94.36%**	**91.15%**	**89.58%**
LSTM_128_lr0005	20	93.09%	90.76%	89.91%
LSTM_64_lr001	19	93.44%	89.71%	90.30%
GRU_128_lr0005	20	91.00%	89.45%	88.61%
RNN_128_lr001	17	90.85%	88.15%	89.00%
RNN_64_lr001	18	88.85%	84.90%	85.48%

Table 1. Configurations ranked by validation accuracy (epochs = epochs run before stopping).

Only RNN_64 and RNN_128 stopped early (after 18 and 17 epochs), and LSTM_64 stopped after 19 epochs; the other three configurations ran for the full 20-epoch limit. Because the configurations change units, dropout and learning rate together, differences between them cannot be attributed to a single factor. The differences among the top four models are also small: with 1,536 evaluation samples, the standard error of an accuracy near 90% is roughly 0.8 percentage points, so the ordering of the leading LSTM and GRU models should not be over-interpreted.

4.2 Training and Validation Error Curves

Figure 2 shows the classification error against epochs for all six configurations. In every case the training error falls steadily. The validation error follows it during the first epochs and then flattens, leaving a gap of a few percentage points; this gap is largest for the vanilla RNN with 64 units, whose validation error stays well above its training error from about the fifth epoch. The GRU and LSTM curves are smoother than the RNN curves.

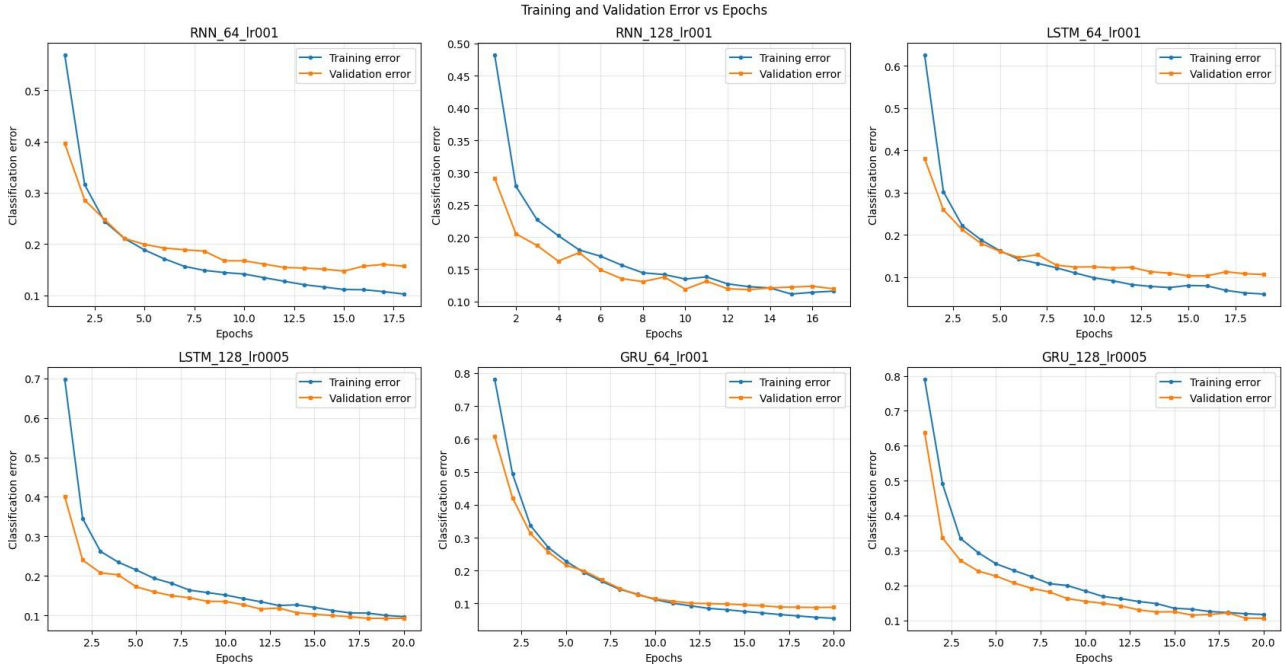


Figure 2. Training and validation error versus epochs for the six configurations.

4.3 Best Configuration

The configuration with the highest validation accuracy is GRU_64_lr001: a GRU with 64 units, no dropout and learning rate 0.001. It reached 94.36% training accuracy, 91.15% validation accuracy and 89.58% test accuracy. Figure 3 shows its error curve.

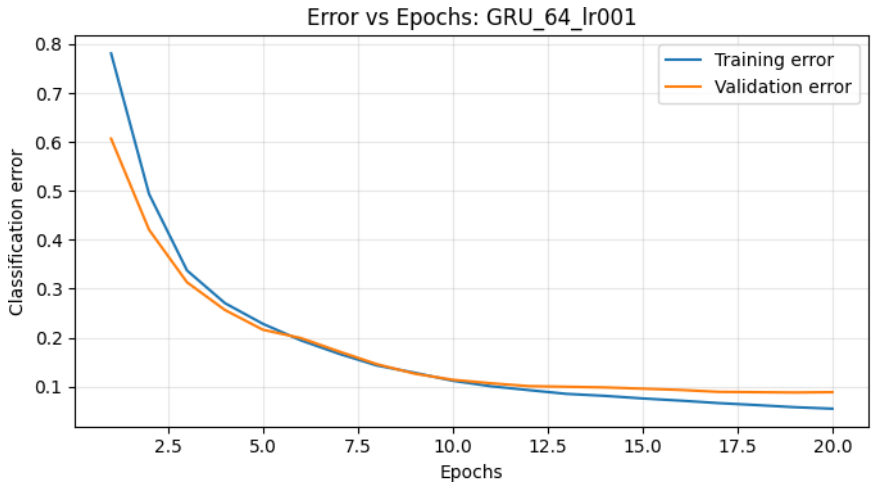


Figure 3. Training and validation error for the selected GRU_64_lr001 model.

4.4 Test-Set Evaluation

On the 1,536 unseen test images the selected model classified 89.58% correctly, which corresponds to 160 misclassified samples. The test accuracy is about 1.6 points below the validation accuracy, consistent with the sampling uncertainty noted above. The per-class results are given below.

Digit	Precision	Recall	F1-score	Support
0	0.8616	0.8954	0.8782	153
1	0.8917	0.9150	0.9032	153
2	0.9178	0.8701	0.8933	154
3	0.9272	0.9091	0.9180	154
4	0.8924	0.9156	0.9038	154
5	0.9080	0.9610	0.9338	154
6	0.8889	0.7792	0.8304	154
7	0.8537	0.9150	0.8833	153
8	0.8598	0.9156	0.8868	154
9	0.9712	0.8824	0.9247	153
Macro average	**0.8972**	**0.8958**	**0.8956**	**1536**

Table 2. Classification report of GRU_64_lr001 on the test set.

4.5 Confusion-Matrix Analysis

Digit 5 is recognized best (recall 96.10%, 148 of 154 correct) and digit 9 has the highest precision (97.12%). Digit 6 is the hardest class, with a recall of only 77.92%: 17 of the 154 test images of digit 6 are predicted as digit 7, and a further six are predicted as digit 8. Digit 0 has the lowest precision (86.16%) because several other classes are occasionally predicted as 0. Other noticeable confusions are digit 4 predicted as digit 5 (8 samples) and digit 2 predicted as digit 8 (6 samples). Most of these pairs have visually similar strokes in the sample images of Figure 1.

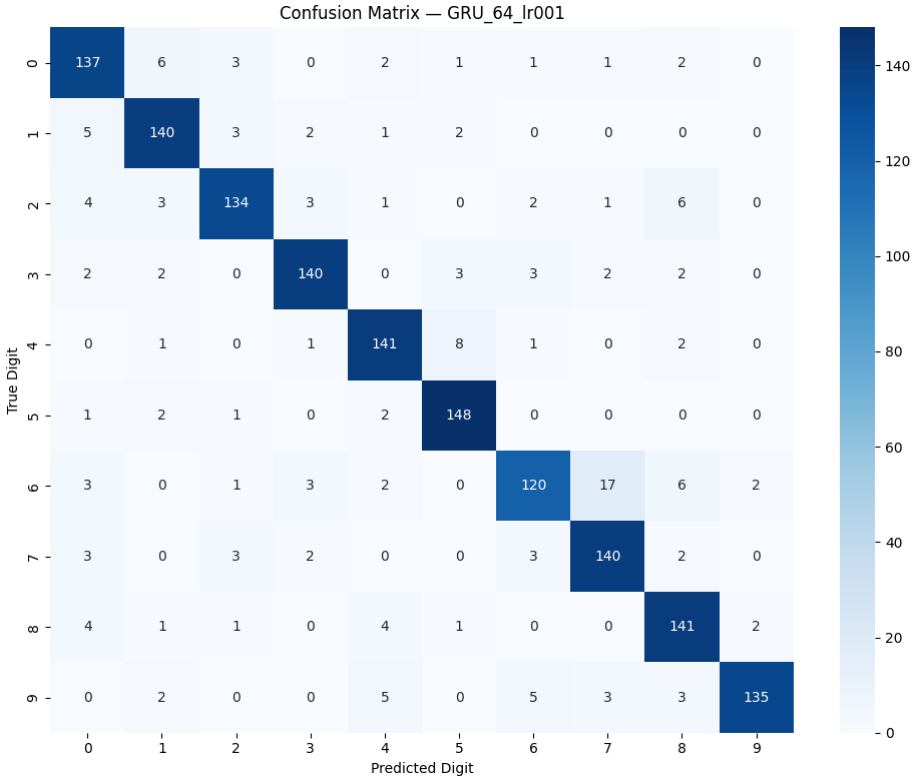


Figure 4. Test confusion matrix for GRU_64_lr001.

4.6 Comparison of RNN, LSTM and GRU

Taking the best configuration of each cell type, the GRU has the highest validation accuracy, the LSTM has the highest test accuracy, and the vanilla RNN is lowest on validation accuracy.

Architecture	Best Validation Acc.	Best Test Acc.
GRU	91.15%	89.58%
LSTM	90.76%	90.30%
RNN	88.15%	89.00%

Table 3. Best validation and best test accuracy for each recurrent architecture.

The maximum validation and maximum test accuracies in Table 3 are taken independently, so they can come from different configurations (for example, the best LSTM test accuracy belongs to LSTM_64_lr001). The gated cells outperform the vanilla RNN on validation accuracy by about 2.6 to 3.0 points, but the test accuracies of the three architectures lie within about 1.3 points of each other, so the evidence for a clear winner on this split is limited.

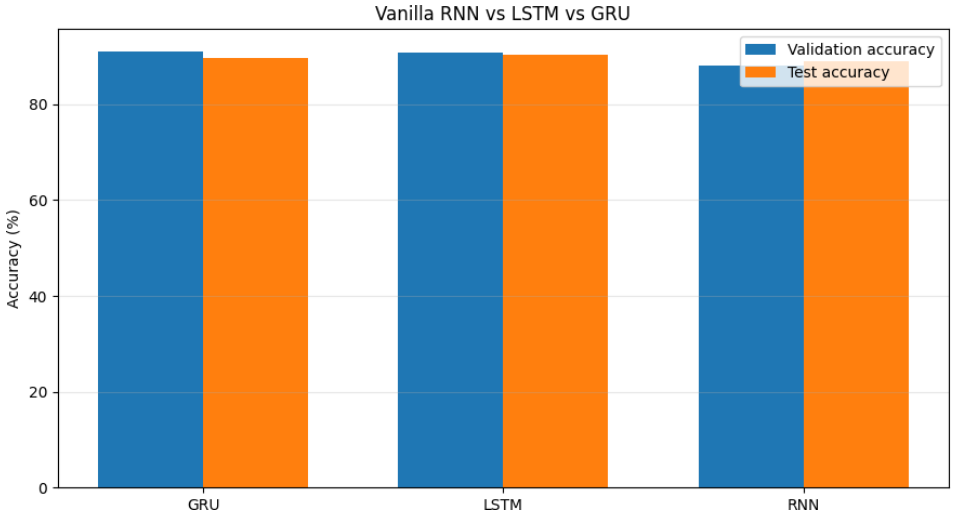


Figure 5. Best validation and test accuracy for vanilla RNN, LSTM and GRU.

4.7 Row-wise Sequential Feature Heatmaps

Figure 6 displays one test image of each digit as the sequence that the recurrent layer actually receives: each row is a time step and each column is one of the 28 pixel features. The heatmaps show that the position of the stroke shifts from row to row, so the network must integrate information across time steps to separate digits whose shapes differ only in how the strokes evolve vertically.

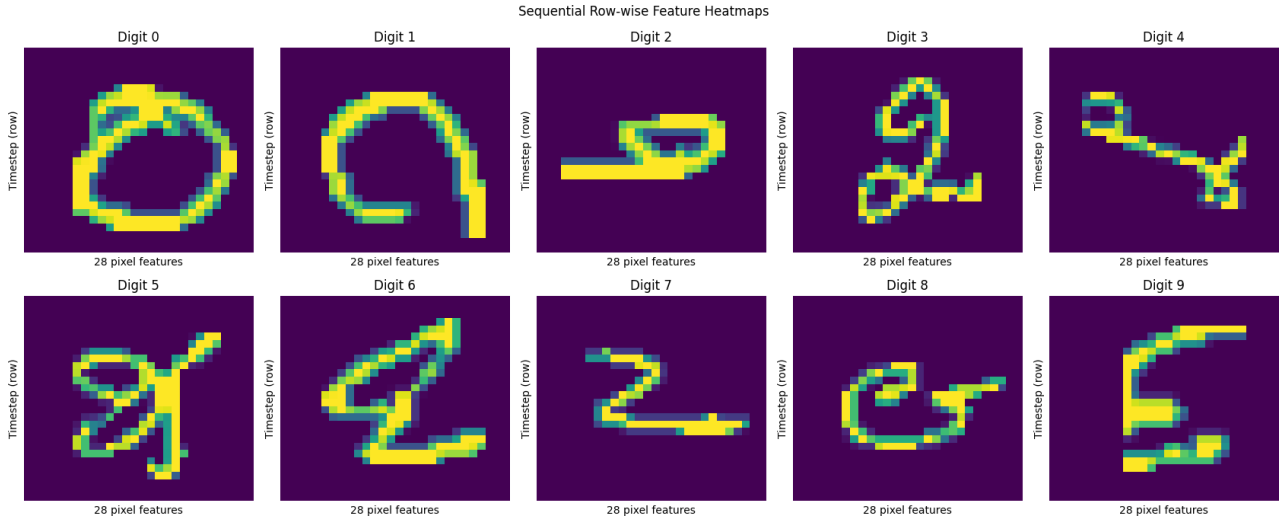


Figure 6. Row-wise sequential feature heatmaps for one test image per digit.

5. Conclusion

The experiment shows that recurrent networks can classify Kannada digits when each image is read as a sequence of 28 rows. The 64-unit GRU was selected by validation accuracy (91.15%) and obtained 89.58% test accuracy, with digit 6 (often confused with digit 7) as the most difficult class. LSTM and GRU models trained more smoothly and reached higher validation accuracy than the vanilla RNN, although the test accuracies of all three architectures are close. Since the configurations differ in several hyperparameters at once and several runs used the

full 20-epoch limit, a longer training budget and a controlled one-factor-at-a-time search would be needed to separate the effects of model size, dropout and learning rate.

TASK 2: Spoken Arabic Digit Recognition with Recurrent Networks

1. Objectives

- Classify spoken Arabic digits (0–9) from variable-length MFCC sequences.
- Handle variable sequence lengths with padding and packed sequences.
- Compare RNN, LSTM and GRU cells with one and two layers and hidden sizes of 32, 64 and 128.
- Train with Adam under a convergence criterion on the average training error.
- Compare training, validation and test accuracy across all 18 configurations.
- Select the best model by validation accuracy and evaluate it with a confusion matrix.
- Examine gradient norms during training and the flow of gradients through time.

2. Dataset and Preprocessing

2.1 Spoken Arabic Digit Dataset

The data come from the files Train_Arabic_Digit.txt and Test_Arabic_Digit.txt. Each utterance is a sequence of frames, and each frame is a vector of 13 Mel-frequency cepstral coefficients (MFCCs); utterances are separated by blank lines. The training file contains 6,600 utterances (660 per digit) and the test file 2,200 utterances (220 per digit). Utterance lengths range from 4 to 93 frames with a mean of 39.9. Class labels are assigned from the position of each utterance, because the files list the digits in consecutive equal-sized blocks.

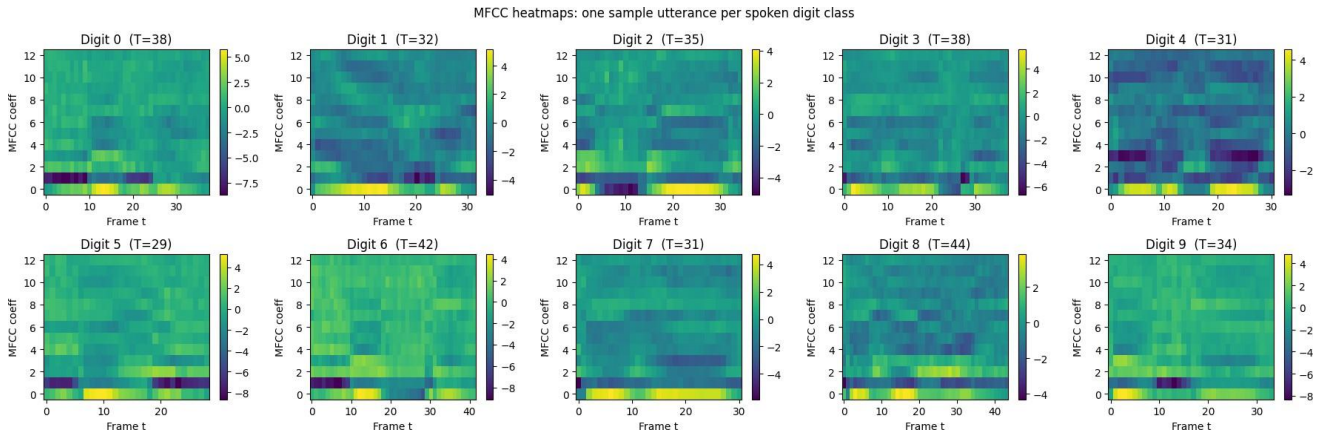


Figure 1. MFCC heatmaps of one sample utterance for each spoken digit class.

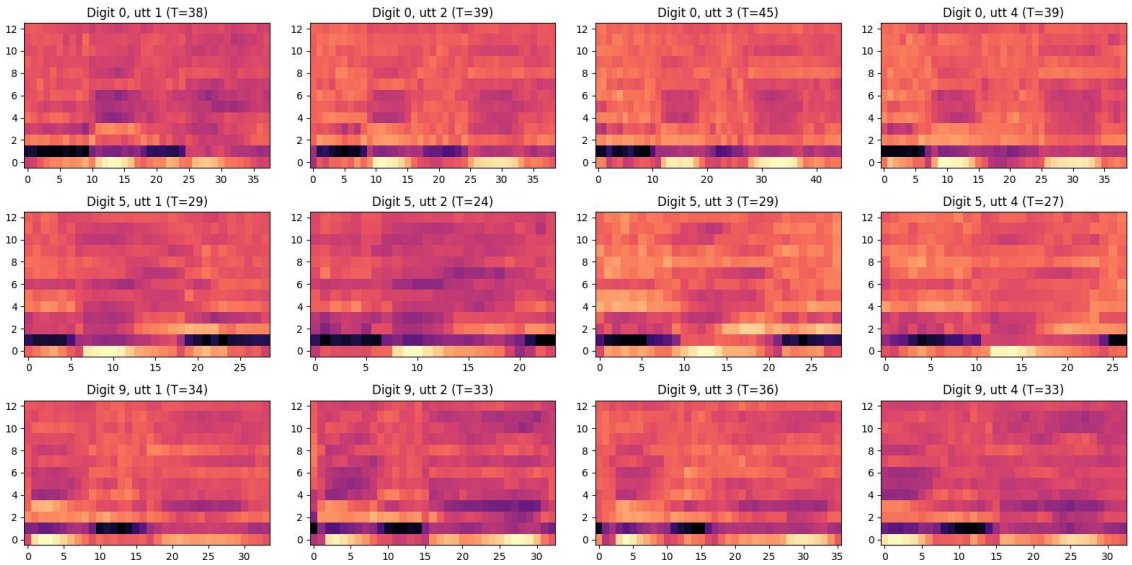


Figure 2. Variability of MFCC patterns across several utterances of digits 0, 5 and 9.

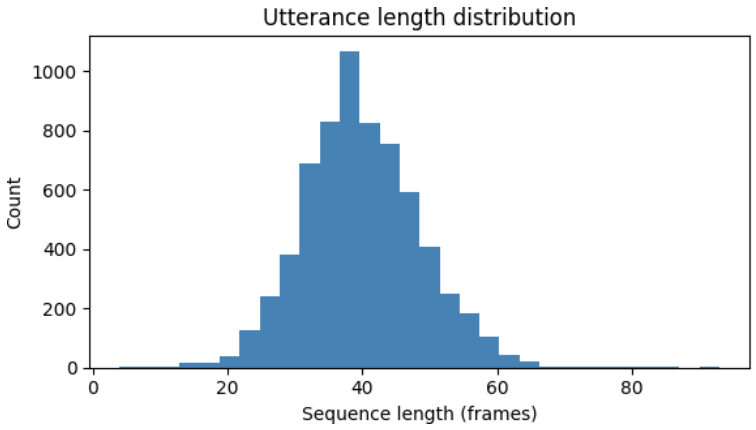


Figure 3. Distribution of utterance lengths in the training file.

2.2 Data Split

The original test file is kept as the held-out test set. The training file was divided with a stratified 90/10 split (random seed 42) into training and validation subsets, so every digit remains balanced.

Subset	Utterances	Per class	Source
Training	5,940	594	Training file (90%)
Validation	660	66	Training file (10%)
Test	2,200	220	Test file

2.3 Input Representation

Each MFCC coefficient was standardized with the mean and standard deviation computed over all frames of the training subset only; the same statistics were applied to the validation and test utterances. Because utterances have different lengths, each mini-batch is zero-padded to its longest sequence and the true lengths are passed to the network, which uses packed sequences so that padded frames are ignored by the recurrent layers.

3. Methodology

3.1 Recurrent Classifier Design

The classifier consists of a recurrent block with a 13-dimensional input followed by a single linear output layer with ten units. The recurrent block is a tanh RNN, an LSTM or a GRU with 1 or 2 layers and hidden size 32, 64 or 128, giving $3 \times 2 \times 3 = 18$ baseline models. Dropout of 0.2 is applied between layers only for the two-layer models. For the baseline models, the hidden state of the last layer at the final valid time step is used as the sequence representation.

Cell	Layers	H = 32	H = 64	H = 128
RNN	1	1,834	5,706	19,594
RNN	2	3,946	14,026	52,618
LSTM	1	6,346	20,874	74,506
LSTM	2	14,794	54,154	206,602
GRU	1	4,842	15,818	56,202
GRU	2	11,178	40,778	155,274

Table 4. Number of trainable parameters of the 18 baseline models.

3.2 Loss Function and Optimization

Cross-entropy (PyTorch CrossEntropyLoss) was used as the loss. All models were trained with Adam, learning rate 0.001, mini-batch size 64, and gradient-norm clipping at 5.0, which guards against exploding gradients in the vanilla RNN. The gradient norm before clipping was recorded at every step for later analysis.

3.3 Initialization and Convergence

The random seed 42 was reset before every run. Training continued until the absolute change in average training loss between consecutive epochs fell below 10^{-4}; this check was applied only from epoch 5 onward, and training was capped at 100 epochs. A run that reaches the cap is reported as not converged.

Setting	Value
Optimizer	Adam (learning rate 0.001)
Batch size	64
Gradient clipping	Norm 5.0
Dropout (2-layer models)	0.2
Convergence tolerance	$\lvert L(k) - L(k-1) \rvert < 10^{-4}$ (from epoch 5)
Maximum epochs	100

4. Experimental Results and Analysis

4.1 Overall Training, Validation and Test Performance

Gated models dominate the table. Fifteen of the 18 runs reach at least 96% validation accuracy, all twelve LSTM and GRU runs exceed 98%, and all of them converged before the epoch cap. The GRU with two layers and 32 hidden units has the highest validation accuracy (99.55%), and it was therefore selected as the best baseline model.

Model	Params	Epochs	Conv.	Train Acc.	Val. Acc.	Test Acc.
GRU L2 H32	**11,178**	**67**	**Yes**	**100.00%**	**99.55%**	**96.27%**
LSTM L2 H64	54,154	28	Yes	100.00%	99.39%	95.41%
GRU L1 H64	15,818	37	Yes	100.00%	99.39%	96.09%
LSTM L2 H128	206,602	24	Yes	100.00%	99.24%	95.45%
GRU L1 H128	56,202	31	Yes	100.00%	99.24%	97.00%
LSTM L1 H64	20,874	38	Yes	100.00%	99.09%	95.45%
LSTM L1 H128	74,506	28	Yes	99.92%	99.09%	95.95%
GRU L2 H64	40,778	19	Yes	99.83%	99.09%	94.73%
GRU L2 H128	155,274	22	Yes	100.00%	99.09%	96.41%
LSTM L2 H32	14,794	36	Yes	99.97%	98.79%	94.64%
GRU L1 H32	4,842	42	Yes	99.88%	98.64%	94.27%
RNN L1 H128	19,594	100	No	99.48%	98.48%	94.64%
LSTM L1 H32	6,346	33	Yes	99.66%	98.03%	94.32%
RNN L2 H128	52,618	100	No	98.84%	96.97%	93.32%
RNN L2 H32	3,946	100	No	98.37%	96.67%	90.36%
RNN L2 H64	14,026	47	Yes	97.32%	95.91%	93.05%
RNN L1 H64	5,706	100	No	98.15%	95.30%	89.09%
RNN L1 H32	1,834	100	No	96.06%	92.42%	85.36%

Table 5. All 18 baseline experiments ranked by validation accuracy (L = layers, H = hidden units).

Every LSTM and GRU model reaches 99.66% to 100% training accuracy, yet the test accuracy stays between 94.3% and 97.0%, so a generalization gap of three to five percentage points remains. The validation accuracy (660 utterances from the same file as the training data) is consistently higher than the test accuracy (2,200 utterances from a separate file), which suggests the validation set is an optimistic estimate of performance on new data. The best test accuracy among the baselines, 97.00%, belongs to GRU L1 H128, not to the selected GRU L2 H32.

4.2 Convergence Behaviour

Cell	Layers	H = 32	H = 64	H = 128
RNN	1	100 (not conv.)	100 (not conv.)	100 (not conv.)
RNN	2	100 (not conv.)	47	100 (not conv.)
LSTM	1	33	38	28
LSTM	2	36	28	24
GRU	1	42	37	31
GRU	2	67	19	22

Table 6. Epochs used by each baseline model.

Five of the six vanilla RNN configurations reached the 100-epoch cap without satisfying the convergence criterion, and their average training loss is noisy rather than smoothly decreasing. The LSTM and GRU models converged in 19 to 67 epochs and reached much smaller final training losses (mostly between 0.0005 and 0.01, against 0.04 to 0.15 for the RNN models). Convergence under this criterion is not by itself a measure of quality, but here the models that converged were also the ones that generalized best.

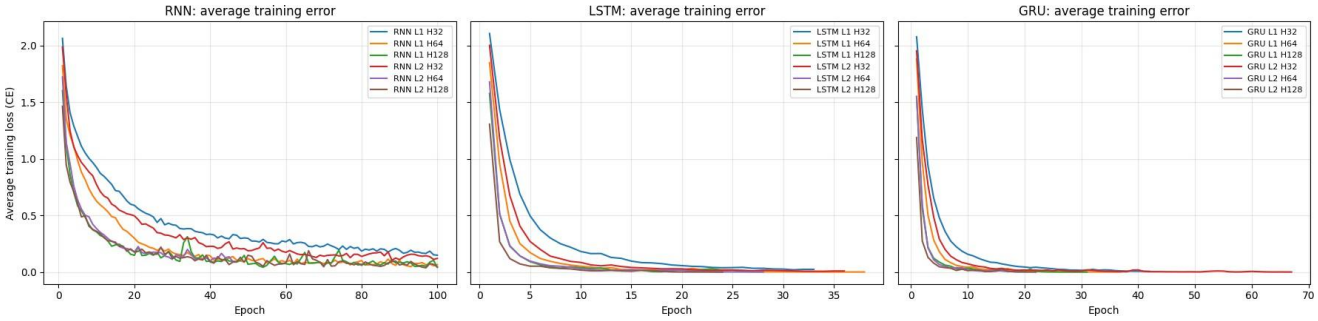


Figure 4. Average training error (cross-entropy) per epoch for each cell type.

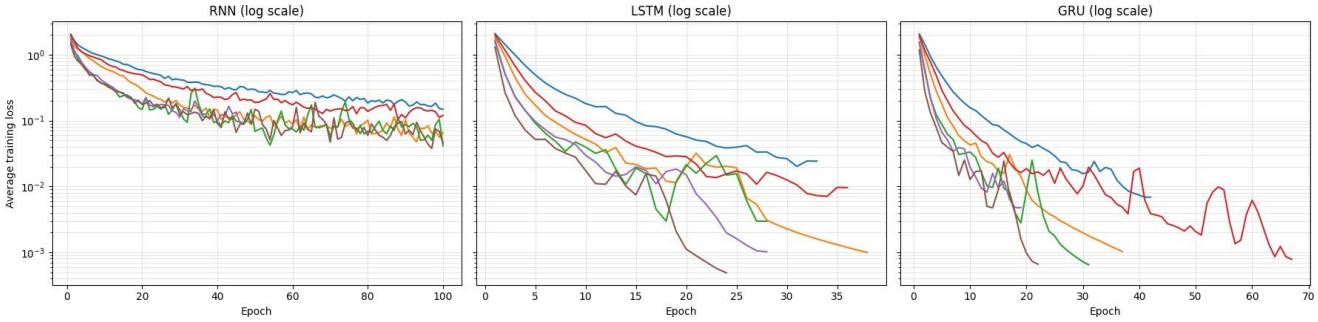


Figure 5. Average training error per epoch on a logarithmic scale.

The logarithmic plot in Figure 5 makes the difference clear: the RNN losses decrease slowly and with large fluctuations, whereas the LSTM and GRU losses drop by several orders of magnitude and then terminate once the change falls below the tolerance.

4.3 Comparison of Recurrent Cell Types

Cell type	Mean Train Acc.	Mean Test Acc.	Mean Epochs
GRU	99.95%	95.80%	36.3
LSTM	99.92%	95.20%	31.2
RNN	98.04%	90.97%	91.2

Table 7. Mean performance of each cell type over its six baseline configurations.

The GRU has the highest average test accuracy (95.80%), the LSTM is close behind (95.20%), and the vanilla RNN is about five points lower (90.97%) and needs about 2.5 to 3 times as many epochs. Gated cells are therefore both easier to train and more accurate in this experiment. The GRU–LSTM difference of 0.6 points is small relative to the variation between configurations.

4.4 Effect of Depth and Width

Cell	Layers	H = 32 (Val / Test)	H = 64 (Val / Test)	H = 128 (Val / Test)
RNN	1	92.42 / 85.36	95.30 / 89.09	98.48 / 94.64
RNN	2	96.67 / 90.36	95.91 / 93.05	96.97 / 93.32
LSTM	1	98.03 / 94.32	99.09 / 95.45	99.09 / 95.95
LSTM	2	98.79 / 94.64	99.39 / 95.41	99.24 / 95.45
GRU	1	98.64 / 94.27	99.39 / 96.09	99.24 / 97.00
GRU	2	99.55 / 96.27	99.09 / 94.73	99.09 / 96.41

Table 8. Validation and test accuracy (%) by layers and hidden size.

For single-layer models, test accuracy increases with the hidden size for all three cell types (RNN 85.36% → 94.64%, LSTM 94.32% → 95.95%, GRU 94.27% → 97.00%). A second layer helps the small vanilla RNNs a lot (for H = 32, the test accuracy rises from 85.36% to 90.36%) but makes little consistent difference for LSTM, and for GRU the effect is mixed.

4.5 Best Standard Model and Test-Set Evaluation

The best baseline model by validation accuracy is GRU L2 H32 (converged after 67 epochs; 100.00% training, 99.55% validation, 96.27% test accuracy). The per-class results and confusion matrix are given below.

Digit	Precision	Recall	F1-score	Support
0	0.9142	0.9682	0.9404	220
1	0.9690	0.9955	0.9821	220
2	0.9641	0.9773	0.9707	220
3	0.9577	0.9273	0.9423	220
4	0.9729	0.9773	0.9751	220
5	0.9234	0.9864	0.9538	220
6	0.9854	0.9182	0.9506	220
7	0.9631	0.9500	0.9565	220
8	0.9904	0.9364	0.9626	220
9	0.9954	0.9909	0.9932	220
Average	**0.9636**	**0.9627**	**0.9627**	**2200**

Table 9. Classification report of GRU L2 H32 on the test set.

Digit 9 is the easiest (F1 0.9932) and digit 1 has the highest recall (99.55%). The weakest results involve digit 0, which has the lowest precision (91.42%), and digit 6, which has the lowest recall (91.82%): 16 of the 220 test utterances of digit 6 are predicted as digit 0. Digit 8 is also confused with digit 3 (9 utterances), and digit 3 is sometimes predicted as digit 7 (6 utterances).

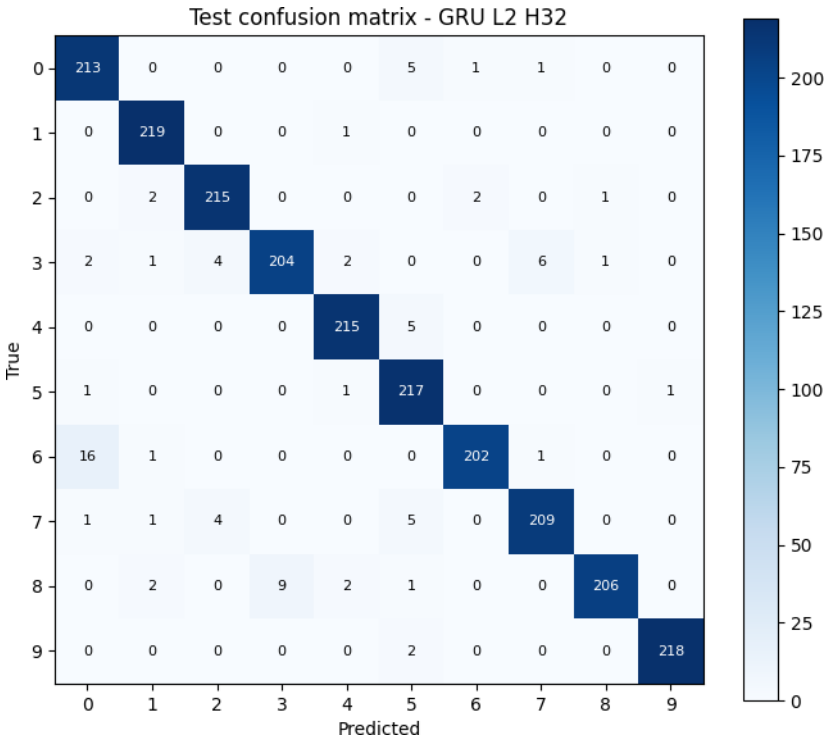


Figure 6. Test confusion matrix for the best baseline model (GRU L2 H32).

4.6 Gradient Norm and Gradient-Flow Analysis

Figure 7 shows the mean gradient norm before clipping during training. The vanilla RNN has large and erratic gradient norms, with spikes of about 30 and 25 against a clipping threshold of 5, whereas the LSTM and GRU gradient norms stay below about 2 and decrease smoothly. This irregular behavior of the RNN is consistent with its noisy loss curve and with the need for gradient clipping.

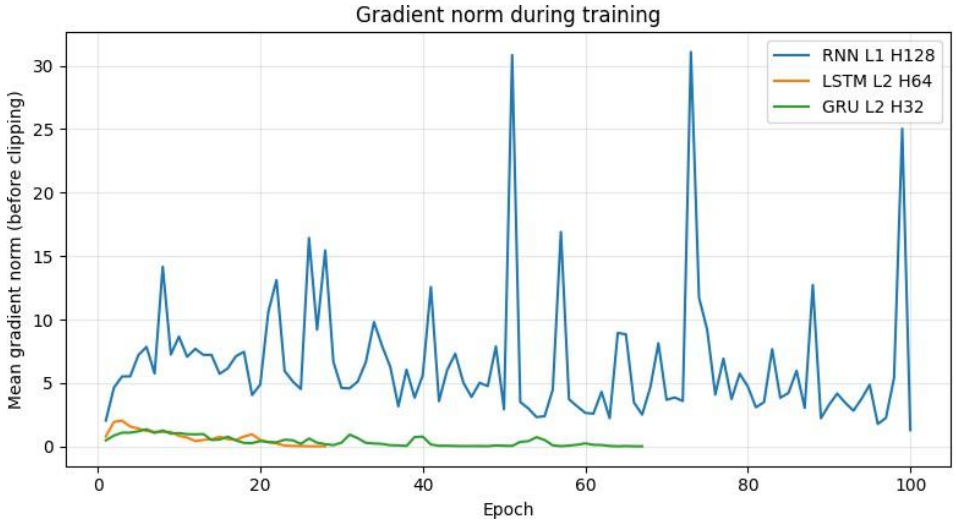


Figure 7. Mean gradient norm (before clipping) per epoch for the best model of each cell type.

Figure 8 shows how strongly the loss depends on input frames at different distances from the last frame (logarithmic scale). For the RNN the gradient is largest at the last frames and falls by roughly an order of magnitude over the 30 steps (from about 10^{-4} to about 1.5×10^{-5}). The

GRU curve is the lowest overall and decays to about 5×10^{-6} at 29 steps back. The LSTM profile is the flattest, staying between roughly 1.5×10^{-5} and 6×10^{-5} over the whole range, which indicates that it keeps the influence of earlier frames better than the other two. These profiles come from models of different sizes and depths, so they illustrate behavior rather than isolate the effect of the cell type; also, the absolute values depend on the scale of the inputs and of the model.

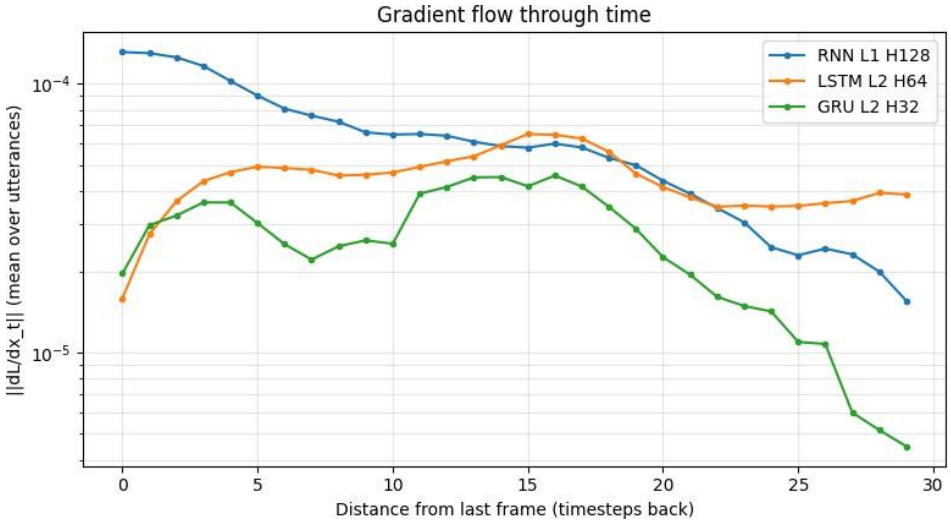


Figure 8. Gradient flow through time: mean input-gradient norm versus distance from the last frame.

4.7 Summary of the Best Model per Cell Type

Cell	Best config	Epochs	Train Acc.	Val. Acc.	Test Acc.	Time (s)
RNN	RNN L1 H128	100	99.48%	98.48%	94.64%	301.5
LSTM	LSTM L2 H64	28	100.00%	99.39%	95.41%	358.5
GRU	GRU L2 H32	67	100.00%	99.55%	96.27%	454.0

Table 10. Best baseline model of each cell type (selected by validation accuracy) and its CPU training time.

5. Bonus: Bidirectional and Attention Extensions

This bonus part goes beyond the required 18 baseline experiments. It tests whether bidirectional processing and attention pooling improve the best baseline model of each cell type, using the same data split, optimizer, loss, gradient clipping and convergence criterion as in Section 3.

5.1 Objectives

- Extend the best baseline of each cell type (RNN L1 H128, LSTM L2 H64 and GRU L2 H32) with bidirectional and attention variants.
- Compare each variant with its baseline using training, validation and test accuracy.
- Select the best overall model by validation accuracy and examine its confusion matrix.

5.2 Methodology

For the best baseline of each cell type (selected by validation accuracy), three variants were trained: a bidirectional model, a model with temporal attention, and a model with both. The

bidirectional models concatenate the final forward and backward hidden states. The attention models replace the last-state representation with a weighted sum of the outputs of all valid frames, where the weights are a softmax over a learned linear score and padded frames are masked out.

Because the baselines were chosen by validation accuracy, the three baselines are RNN L1 H128, LSTM L2 H64 and GRU L2 H32. Each of them gives three variants, Bi, Attn and Bi Attn, for $3 \times 3 = 9$ additional models. The bidirectional variants roughly double the number of recurrent parameters, while the attention layer adds only a single linear scoring layer.

5.3 Results

The extensions were applied to the best baseline of each cell type: RNN L1 H128, LSTM L2 H64 and GRU L2 H32. The table compares each variant with its baseline.

Model	Params	Epochs	Train Acc.	Val. Acc.	Test Acc.
RNN L1 H128 (baseline)	**19,594**	**100**	**99.48%**	**98.48%**	**94.64%**
RNN L1 H128 Bi	39,178	32	99.61%	98.48%	96.45%
RNN L1 H128 Attn	19,723	60	100.00%	99.09%	97.00%
RNN L1 H128 Bi Attn	39,435	40	100.00%	99.39%	97.14%
LSTM L2 H64 (baseline)	**54,154**	**28**	**100.00%**	**99.39%**	**95.41%**
LSTM L2 H64 Bi	141,066	16	100.00%	99.39%	96.14%
LSTM L2 H64 Attn	54,219	37	100.00%	99.39%	97.14%
LSTM L2 H64 Bi Attn	141,195	21	99.87%	99.39%	96.36%
GRU L2 H32 (baseline)	**11,178**	**67**	**100.00%**	**99.55%**	**96.27%**
GRU L2 H32 Bi	28,490	29	100.00%	99.70%	97.55%
GRU L2 H32 Attn	11,211	44	100.00%	99.39%	96.95%
GRU L2 H32 Bi Attn	28,555	26	100.00%	99.55%	98.00%

Table 11. Baseline versus bidirectional (Bi) and attention (Attn) variants (baseline rows in bold).

Every extension improves the test accuracy of its baseline. The gain is largest for the vanilla RNN (94.64% → 97.14% with Bi + Attn), for which attention pooling alone already gives 97.00%; it is smaller for the LSTM (95.41% → 97.14% with attention) and the GRU (96.27% → 98.00% with Bi + Attn). All extended models converged, including the RNN variants whose baseline did not, and most variants needed fewer epochs than their baselines (the exception is LSTM L2 H64 Attn, with 37 epochs against 28). The best overall model by validation accuracy is GRU L2 H32 Bi (99.70% validation, 97.55% test). The highest test accuracy overall, 98.00%, is obtained by GRU L2 H32 Bi Attn, but its validation accuracy (99.55%) is lower, so it is not the validation-selected model. Each variant was trained once with a single seed, so differences of a few tenths of a point between variants are within the range of run-to-run variation.

5.4 Training Error of the Extended Models

Figure 9 shows the average training loss of the nine extended models. All of them decrease quickly in the first few epochs and then approach zero, although a few variants show small bumps later in training (for example, one of the attention variants around epoch 20–30).

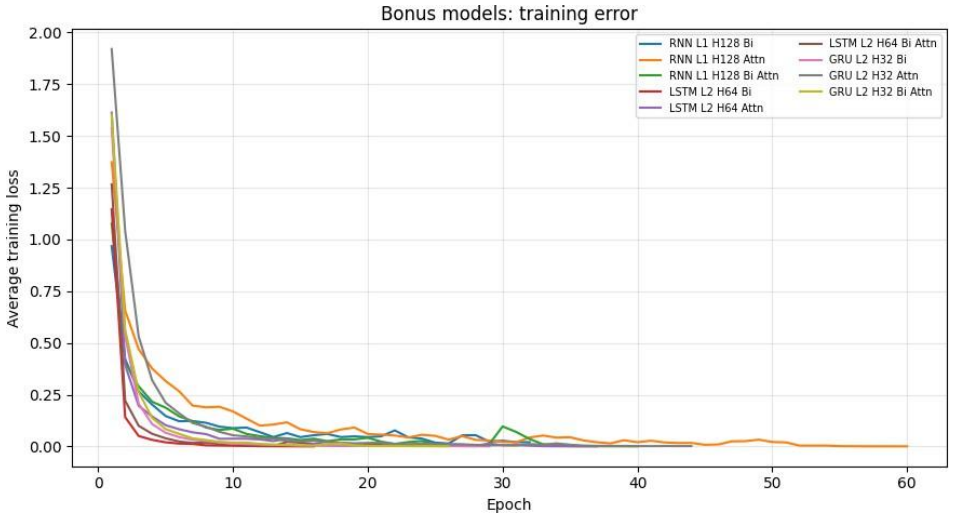


Figure 9. Training error of the bidirectional and attention variants.

5.5 Best Overall Model

The confusion matrix of the selected extended model shows the same dominant error pattern as the baseline, but with fewer mistakes overall (54 errors against 82 on the 2,200 test utterances): digit 6 is still predicted as digit 0 for 13 utterances, and digit 8 is predicted as digit 3 for 6 utterances.

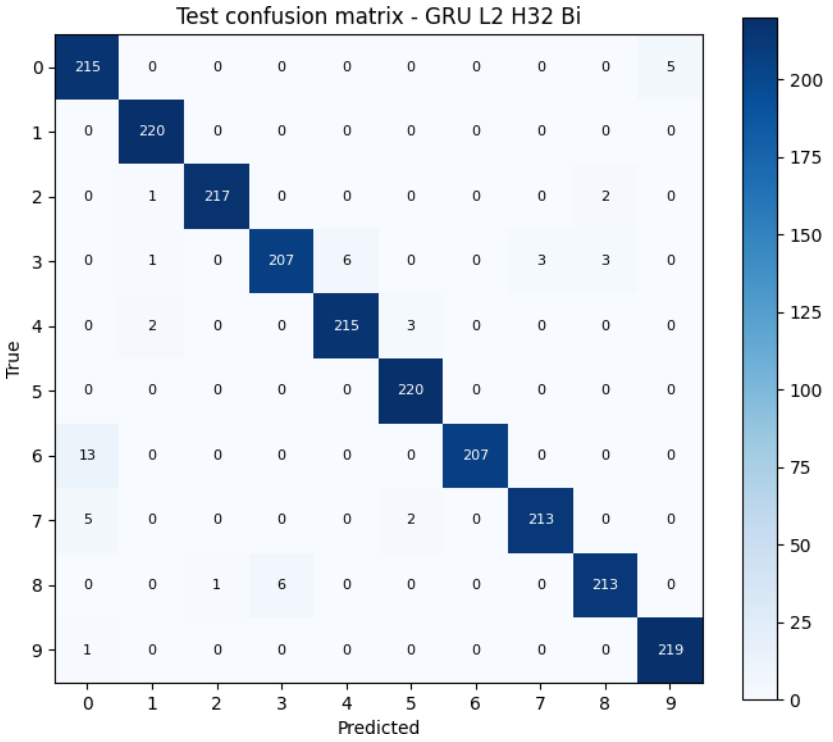


Figure 10. Test confusion matrix for the best overall model by validation accuracy (GRU L2 H32 Bi).

6. Conclusion

This experiment compared RNN, LSTM and GRU classifiers for spoken Arabic digits on padded and packed MFCC sequences. Gated cells clearly outperformed the vanilla RNN: the

average test accuracy was 95.80% for GRU, 95.20% for LSTM and 90.97% for RNN, and almost all gated models converged within the epoch limit while five of six RNN models did not. The best baseline by validation accuracy was a two-layer GRU with 32 hidden units (99.55% validation, 96.27% test). In the bonus part, bidirectional processing and attention pooling improved the test accuracy of every baseline, with the bidirectional two-layer GRU selected as the best overall model by validation accuracy (99.70% validation, 97.55% test). The gradient diagnostics support these findings by showing erratic, spiking gradient norms for the vanilla RNN and smooth ones for LSTM and GRU. The persistent gap between near-perfect training accuracy and 94% to 98% test accuracy suggests that regularization or data augmentation, and repeated runs with several seeds, would be the next steps for a more reliable comparison.